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1. Introduction

1.1. Project Context

The focus of this project is on the development of a robust Inventory Management web application for a candy company. The system is designed to enable inventory to be tracked and managed by multiple users, with related tasks, such as generating reports and issuing invoices, being completed depending on the user's role. The management of inventory within the company is intended to be streamlined by this system.

1.2. Software Engineering Principles Applied

Agile methodology, Scrum, was applied for iterative development. A total of two sprints were conducted. This principle ensured the project was broken into smaller, manageable tasks, allowing features to be developed incrementally. A clear timeline for task completion was also provided, ensuring the project remained on track.

2. Project Scope

2.1. Design

A) Architectural Design: The system will have a web-based front end that will be developed using HTML, CSS, and PHP, with a MySQL database as the backend. The user interface will be provided by the front end, while information will be stored in the database and managed or retrieved using PHP.

B) Database Design: A relational database (MySQL) will be used for storing user details, inventory information, and over-the-counter sales.

C) Interface Design: A responsive user interface will be designed for managers, administrators, and sales clerks.

D) Component Design:

1. Inventory Tracking Component (for viewing, adding, updating and deleting stock)
2. Invoice Issuing Component (issue invoices for sales)

2.2. Implementation

Development will occur incrementally, with components being improved as feedback is received. Inventory management will be implemented first, followed by login and user details, and invoice issuing.

2.3. Testing

1. Unit Testing: Individual components, such as adding and updating stock, will be frequently tested.
2. Security Testing (OWASP ZAP): Identify potential vulnerabilities, including input validation, authentication, & session management.
3. Accessibility Testing (WAVE Evaluation): Ensure application is user-friendly & meets user experience expectations.
4. User Acceptance Testing: Various positive and negative tests will be run by managers and employees to ensure usability and functionality needs are met by the system.
5. Performance Testing: Ensure that the system can handle peak loads, such as recording user details and inventory data.

3. Functional Requirements

- Enable inventory monitoring
- Allow stock addition and updates
- Generate management reports
- Issue over-the-counter sales invoices
- Support multiple branches with a responsive web interface

4. Non-Functional Requirements

ID	Theme	Description	Priority (MoSCoW)
NFR01	Availability	Maintain 99.9% uptime with backups for continuous accessibility.	Must
NFR02	Performance	Response time for actions (e.g., submitting forms, logging in) should be under 3 seconds.	Must
NFR03	Legislative Requirements	Data security & privacy must comply with GDPR.	Must
NFR04	Security requirements	Restrict inventory management access to authorised personnel only.	Must